Quantification: 872_wt_hp_z04
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	20
interior lacuna count	17
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1972.35	1593	1094.63	px^2
major_axis_length_px	lacuna shape	95.4116	76.9066	60.9545	px
minor_axis_length_px	lacuna shape	31.4677	29.411	9.6687	px
aspect_ratio	lacuna shape	3.3434	2.5083	2.8083	unitless
eccentricity	lacuna shape	0.8849	0.9171	0.1051	unitless
solidity	lacuna shape	0.7387	0.7622	0.0886	unitless
roots_count	headline	12.2941	12	6.2025	count
ring_length_r30_px	headline	518.647	499	262.695	px
ring_length_r60_px	reported	1060.94	1102	623.672	px
owned_length_px	ownership-dependent	1953.11	1919.47	1665.13	px
edge_count	ownership-dependent	128.765	122	111.584	count
mean_edge_length_px	ownership-dependent	14.4474	15.1436	4.1589	px
perimeter_px	normalised	291.755	256.208	149.922	px
ring_area_r30_px2	normalised	8951.12	8953	3460.37	px^2
ring_area_r60_px2	normalised	19286.5	20267	6525.92	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9957	1	0.0177	unitless
ring_density_r30	normalised	0.0581372	0.0603675	0.0247068	px^-1
ring_density_r60	normalised	0.0542278	0.051861	0.0266922	px^-1
roots_per_100px_perimeter	normalised	4.5225	4.8447	2.3094	per 100 px
ring_attached_length_r30_px	network v2	488	459	260.518	px
ring_attached_length_r60_px	network v2	976.412	1011	629.828	px
ring_length_w_r30_px	network v2	592.677	565.132	300.927	px
ring_length_w_r60_px	network v2	1208.83	1243.99	714.645	px
sholl_crossings_r10	network v2	11.5882	11	6.3744	count
sholl_crossings_r20	network v2	11.4118	13	6.9557	count
sholl_crossings_r30	network v2	14.5294	13	8.4714	count
loop_count	loops, report only	19.2941	8	23.8006	count
loop_edge_share	loops, report only	0.3796	0.3629	0.2507	share
ring_width_mean_r30_px	width	7.087	7.1622	0.4148	px

Per field
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measure	value	units
analysed_area_px2	1.00684e+06	px^2
total_skeleton_length_px	38704	px
canalicular_length_density_per_px	0.0384409	px^-1
junction_count	1400	count
junction_density_per_px2	0.00139048	px^-2
lacuna_count	20	count

lacunae_per_px2	1.986e-05	px^-2
skeleton_component_count	144	count
mean_component_length_px	268.778	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0435637	px^-1
field_density_without_flagged_per_px	0.0430632	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	80	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0384409	px^-1
field_length_density_with_unattached_per_px	0.0385204	px^-1
loop_count	475	count
loop_edge_share	0.594	share
edge_count	2744	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6707	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	36714	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3185	293.107	22.7028	12.9106	0.997	0.5399	-1.5651	143.896	232.375	False
2	2077	144.059	25.2241	5.7112	0.9846	0.6281	1.3769	151.947	448.976	False
3	4256	158.774	41.7686	3.8013	0.9648	0.6782	-1.4497	152.645	678.274	False
4	1429	52.067	37.8414	1.3759	0.6869	0.8279	-0.6051	154.502	593.396	False
5	1593	70.4006	31.8422	2.2109	0.8919	0.7622	1.3948	164.237	538.472	False
6	922	97.7946	15.9727	6.1226	0.9866	0.6272	1.3884	181.894	939.311	False
7	3025	100.943	44.409	2.273	0.898	0.65	-1.4641	211.119	126.158	False
8	2144	87.4647	34.0354	2.5698	0.9212	0.7528	-1.2557	226.871	777.66	False
9	2841	103.213	37.4578	2.7554	0.9318	0.8351	-1.3658	276.205	420.61	False
10	497	30.0795	23.8394	1.2618	0.6098	0.7827	-0.6524	264.441	629.968	False
11	3125	68.6713	60.8024	1.1294	0.4648	0.909	0.622	332.681	995.111	True
12	3954	114.494	48.2385	2.3735	0.9069	0.7517	-1.4049	393.248	55.5005	True
13	3573	94.8523	52.0807	1.8213	0.8358	0.8284	-1.4851	395.903	280.417	False
14	1409	76.4593	28.6406	2.6696	0.9272	0.7683	-1.5183	392.216	831.581	False
15	2226	76.9066	39.7827	1.9332	0.8558	0.7987	-1.558	506.567	185.868	False
16	1503	75.7312	26.0285	2.9095	0.9391	0.8628	-1.3051	568.045	702.943	False
17	1123	61.8357	30.8107	2.007	0.867	0.6785	-1.3397	588.985	20.1576	True
18	628	38.8866	23.9678	1.6225	0.7875	0.7235	-1.4785	619.033	332.451	False
19	1563	73.771	29.411	2.5083	0.9171	0.717	-1.5252	651.639	225.151	False
20	659	47.4878	19.9461	2.3808	0.9075	0.7744	1.5081	843.693	521.331	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	19	668	1102	1659.8	122	13.6049	6.5894
2	8	344	557	549.333	36	15.2593	6.6102
3	11	462	540	587.316	45	13.0515	7.3547
4	4	181	210	208.782	17	12.2813	7.1239
5	5	188	319	271.238	15	18.0825	7.012
6	12	302	512	347.764	20	17.3882	6.5629
7	23	840	1399	1919.47	115	16.6911	7.2004
8	18	756	1370	1877.81	124	15.1436	7.2133
9	14	754	1743	2169.29	139	15.6064	7.206
10	9	449	1046	3143.42	192	16.372	7.8128
11	13	358	847	1304.12	86	15.1641	7.5825
12	17	607	1362	1370.58	94	14.5807	7.3359
13	17	874	2140	4594.31	305	15.0633	7.846
14	12	409	1137	2198.56	126	17.4489	6.4577
15	20	783	1870	2108.23	171	12.3288	7.3483
16	9	506	1063	2047.17	113	18.1165	6.8757
17	12	482	1240	2316.1	155	14.9426	6.8827
18	11	499	1252	6152.76	413	14.8977	6.8692
19	17	802	1761	3367.6	236	14.2695	7.3103
20	0	0	15	0	0	0	

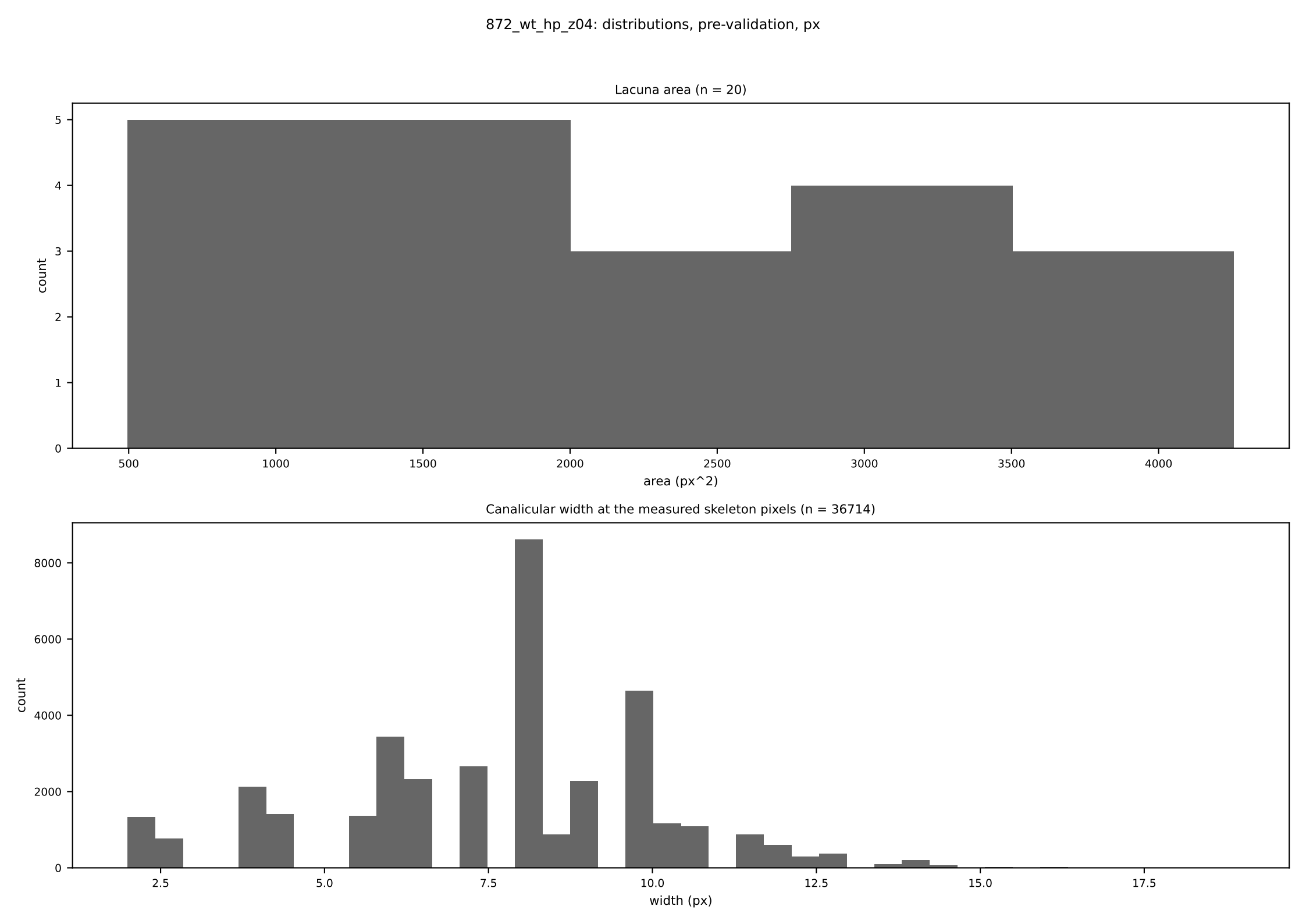
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	728.713	18655	35000	1	1	0.0358081	0.0314857	2.6073
2	364.67	8953	15872	1	1	0.0384229	0.0350932	2.1938
3	505.132	12229	20274	1	1	0.0377791	0.0266351	2.1776
4	197.238	3616	6552	1	1	0.0500553	0.0320513	2.028
5	234.723	4437	8107	1	1	0.042371	0.0393487	2.1302
6	237.723	8984	20267	1	0.9271	0.0336153	0.0252627	5.0479
7	425.345	10604	20758	1	1	0.0792154	0.0673957	5.4074
8	273.321	9813	18631	1	1	0.0770407	0.0735334	6.5857
9	288.978	10581	24146	1	1	0.0712598	0.0721859	4.8447
10	117.054	5613	13620	1	1	0.0799929	0.0767988	7.6888
11	236.551	6707	15609	0.711	0.6585	0.0533771	0.0542636	5.4956
12	432.996	10444	22425	0.8508	0.7745	0.0581195	0.0607358	3.9261
13	315.191	11229	25289	1	1	0.0778342	0.0846218	5.3936
14	256.208	8359	21924	1	1	0.0489293	0.051861	4.6837
15	273.764	9314	21493	1	1	0.084067	0.0870051	7.3055
16	204.445	8382	22359	1	1	0.0603675	0.0475424	4.4022
17	216.752	5820	13951	0.7153	0.655	0.0828179	0.0888825	5.5363
18	148.61	6134	15204	1	1	0.0813499	0.0823467	7.4019
19	243.386	8889	20037	1	1	0.0902239	0.0878874	6.9848
20	145.338	6377	18338	1	1	0	0.00081797	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	617	1029	759.014	1243.99	20	13	11
2	259	533	388.037	639.76	6	4	10
3	455	537	536.274	629.115	11	8	10
4	181	208	206.711	238.439	3	3	4
5	162	225	216.095	361.593	5	2	5
6	270	332	339.007	575.345	7	6	9
7	809	1173	968.737	1605.09	22	15	16
8	679	1317	866.725	1564.11	17	16	25

9	732	1683	863.654	1997.43	18	19	22
10	448	1011	530.588	1196.31	9	13	11
11	308	675	399.635	964.708	7	12	12
12	564	1127	706.683	1562.71	14	14	14
13	854	2137	998.767	2464.55	19	25	31
14	342	940	458.605	1271.38	9	12	13
15	779	1853	884.441	2140.3	14	17	27
16	459	740	574.132	1183.22	11	8	15
17	475	1239	544.676	1418.12	12	12	14
18	466	1147	565.132	1416.15	10	14	20
19	784	1734	919.583	2005.28	16	19	18
20	0	0	0	18.0711	0	0	0


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 20)
count
area (px^2)
0
1
2
3
4
5
500
1000
1500
2000
2500
3000
3500
4000
Canalicular width at the measured skeleton pixels (n = 36714)
count
width (px)
0
2000
4000
6000
8000
2.5
5.0
7.5
10.0
12.5
15.0
17.5